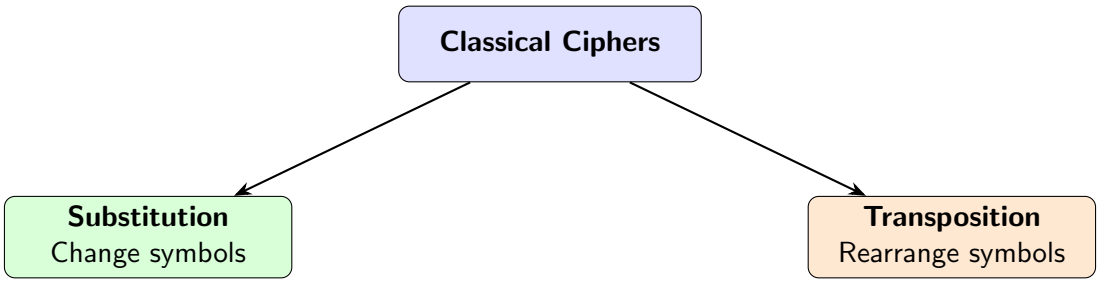
# Introduction to Cryptography

## Lecture 10: Classical Cipher Problems, Rotor Machines and Steganography

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# Review: Classical Cipher Families




**Key Idea**

Substitution changes the symbols. Transposition changes the positions of symbols.

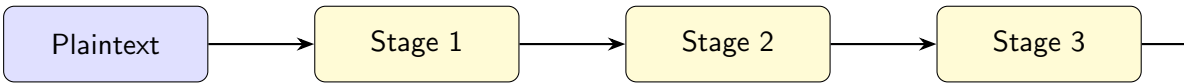
# Why Multiple Stages?

- Simple substitution ciphers preserve language statistics.
- Simple transposition ciphers preserve letter frequencies.
- A single stage is usually easy to analyze.
- Multiple stages create a more complex transformation.
- More stages make the mapping harder to reconstruct.

**Observation**

A double transposition is harder to cryptanalyze than a single transposition. Similarly, multiple substitution stages improve resistance to simple analysis.

# Multiple Stages of Encryption




**Classical Example**

Rotor machines repeatedly apply changing substitution mappings to plaintext letters.

# Rotor Machines

**Definition**

A rotor machine is an electromechanical cipher machine that uses rotating wired cylinders to implement a changing sequence of substitution alphabets.

- Each rotor has 26 input contacts and 26 output contacts.
- Internal wiring maps each input contact to one output contact.
- After a keypress, one or more rotors rotate.
- Rotation changes the substitution alphabet.
- Multiple rotors produce a long-period polyalphabetic substitution.

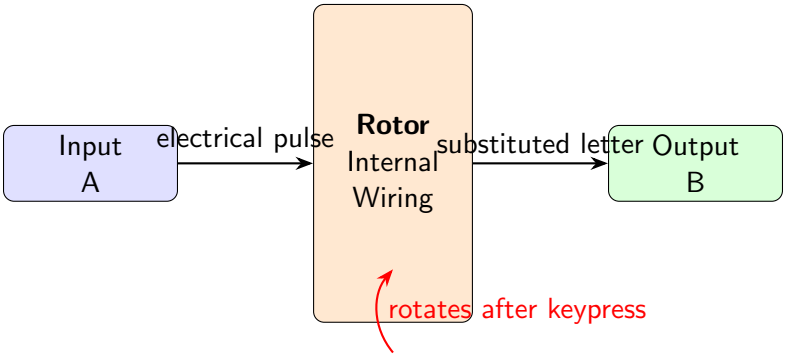
# Historical Note

- Rotor machines were important before modern block ciphers.
- They implemented multiple stages of substitution mechanically.
- The German Enigma and Japanese Purple machines used rotor-machine principles.
- The cryptanalysis of such machines played a major role during World War II.
- Rotor machines influenced the design thinking behind later ciphers such as DES.

**Important**

Rotor machines show how repeated simple substitutions can create a much more complex cipher.

# Single Rotor as a Substitution Cipher




**Concept**

One rotor behaves like a monoalphabetic substitution cipher at a fixed position.

# Effect of Rotor Rotation

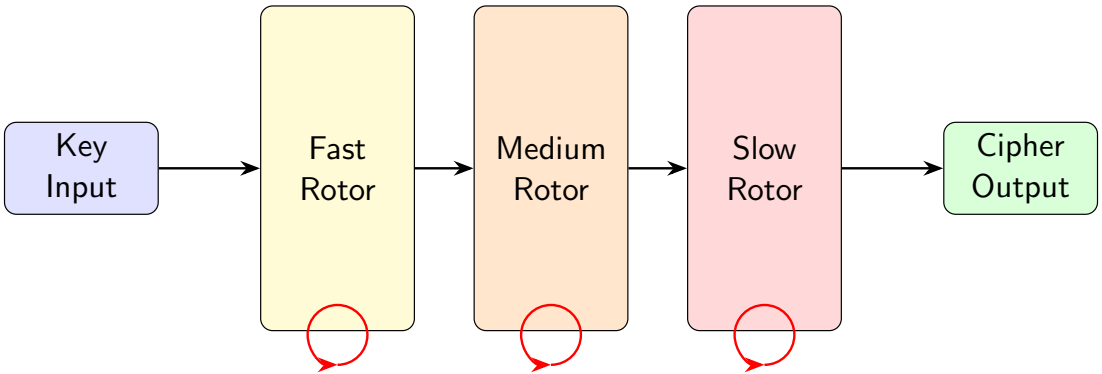
- When the rotor rotates, the input-output wiring alignment changes.
- The same plaintext letter may encrypt to a different ciphertext letter at different times.
- A single rotor has 26 possible positions.
- After 26 keypresses, the rotor returns to its original position.

**Period of One Rotor**

$$26$$

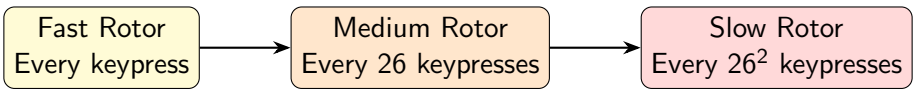
A single rotor creates a polyalphabetic substitution with period 26.

# Three-Rotor Machine: Conceptual Diagram



# Odometer-Like Rotation

- The fast rotor moves one position after every keypress.
- After the fast rotor completes one full rotation, the medium rotor moves one position.
- After the medium rotor completes one full rotation, the slow rotor moves one position.
- This resembles the operation of an odometer.



# Period of Rotor Machines

**Period**

The period is the number of letters encrypted before the rotor machine returns to the same configuration.

For a machine with $r$ rotors, each having 26 positions:

$$\text{Period} = 26^r$$

- 1 rotor: 26
- 2 rotors: $26^2 = 676$
- 3 rotors: $26^3 = 17,576$
- 4 rotors: $26^4 = 456,976$
- 5 rotors: $26^5 = 11,881,376$

# Numerical 1: Period of a Three-Rotor Machine

Find the period of a rotor machine with 3 rotors.

$$\text{Period} = 26^r$$

$$r = 3$$

$$\text{Period} = 26^3 = 26 \times 26 \times 26$$

$$\text{Period} = 17,576$$

**Answer**

The machine uses 17,576 different rotor positions before repeating.

# Numerical 2: Period of a Five-Rotor Machine

Find the period of a machine with 5 rotors.

$$\text{Period} = 26^5$$

$$26^5 = 26 \times 26 \times 26 \times 26 \times 26$$

$$26^5 = 11,881,376$$

**Answer**

A five-rotor machine repeats after 11,881,376 letters.

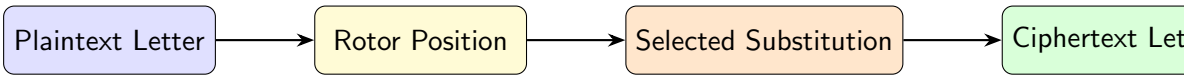
# Why Multiple Rotors Improve Security

- A single rotor has a short period of 26.
- Multiple rotors greatly increase the period.
- The same plaintext letter can map to many different ciphertext letters.
- Frequency analysis becomes less effective.
- The substitution alphabet changes continuously.

**Core Idea**

Rotor machines implement many monoalphabetic substitutions in sequence, controlled by rotor positions.

# Rotor Machine as Polyalphabetic Substitution




**Meaning**

The rotor position determines which substitution alphabet is used for a given plaintext character.

# Cryptanalysis Observation

- Rotor machines reduce the usefulness of simple letter-frequency analysis.
- Long periods make repeated patterns harder to detect.
- However, rotor machines are not automatically unbreakable.
- Weak operating procedures, repeated message keys, and known plaintext can help cryptanalysis.
- Security depends on the machine design, rotor wiring, rotor order, starting positions, and operating discipline.

# From Rotor Machines to Modern Ciphers

- Rotor machines demonstrated the power of repeated transformations.
- Modern block ciphers also use multiple rounds.
- Each round applies substitution and permutation-like operations.
- DES and AES follow this general design philosophy in electronic form.

**Connection**

Rotor machines are historical predecessors to modern iterative encryption algorithms.

# Classical Cipher Problems

Common problems in classical ciphers include:

- Small key space
- Preserved language statistics
- Repeated patterns
- Weak key distribution
- Manual encryption errors
- Vulnerability to known-plaintext attacks

**Lesson**

A cipher must hide both message content and exploitable structure.

# Cryptography vs Steganography

## Cryptography

Transforms the message so that outsiders cannot understand it.

## Steganography

Hides the existence of the message itself.

## Simple Difference

Cryptography says: "You may see the message, but you cannot read it."
Steganography says: "You should not even know that a message exists."

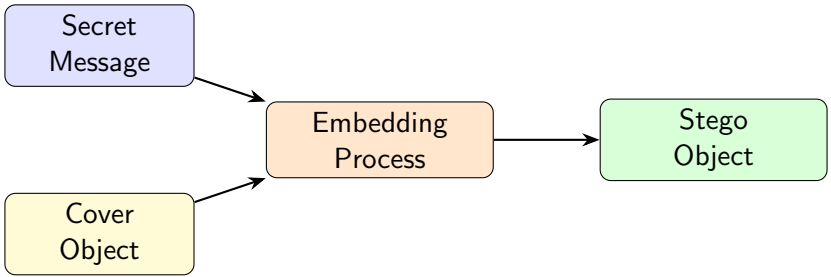
# What is Steganography?

**Definition**

Steganography is the technique of concealing a secret message inside an apparently harmless cover medium.

- The cover may be text, image, audio, video, or network traffic.
- The hidden message should not attract attention.
- The goal is concealment of existence rather than only secrecy of content.

# Steganography Model




**Stego Object**

The output looks ordinary but secretly contains hidden information.

# Classical Steganography Techniques

Historical examples include:

- Character marking
- Invisible ink
- Pin punctures
- Typewriter correction ribbon
- Hidden acrostic messages
- Messages embedded in ordinary-looking text

**Text Example**

The first letter of each word or sentence may spell out a hidden message.

# Character Marking

**Idea**

Selected letters in printed or typed text are marked subtly.

- Marks are not visible under normal reading conditions.
- They may be visible under bright light or at a special angle.
- Marked letters together reveal the hidden message.

**Limitation**

It is slow to create and easy to expose if the method is suspected.

# Invisible Ink

**Idea**

A message is written using a substance that is not visible until heat or chemicals are applied.

- Historically used in espionage.
- The cover paper looks normal.
- The message appears only when treated correctly.

# Pin Punctures and Correction Ribbon

## Pin Punctures

Tiny holes are made near selected letters. The holes are visible only when held against light.

## Correction Ribbon

Typing with correction ribbon between normal lines creates marks visible under strong light.

# Digital Steganography

Modern steganography hides data inside digital media such as:

- Images
- Audio files
- Videos
- Documents
- Network packets
- File metadata

**Common Idea**

Modify parts of the digital object in a way that humans do not easily notice.

# Image Steganography: LSB Method

## Least Significant Bit Method

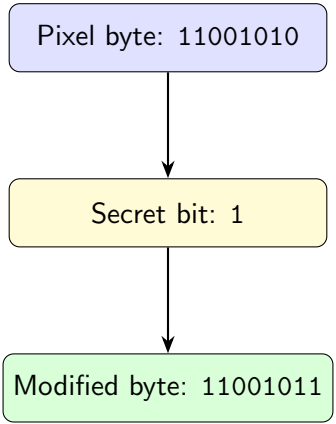
In an image, small changes to the least significant bits of pixel values often produce little visible change.

- Each pixel stores color information.
- The least significant bit has the smallest visual impact.
- Secret message bits can be embedded into these positions.

## Example

Changing RGB value 11001010 to 11001011 may be visually unnoticeable.

# LSB Embedding Illustration




**Observation**

The binary value changes slightly, but the visible pixel color may appear the same.

# Numerical 3: Image Capacity

An image has $2048 \times 3072$ pixels. If 1 bit is hidden in each pixel, find the capacity.

$$\text{Pixels} = 2048 \times 3072 = 6,291,456$$

$$\text{Hidden bits} = 6,291,456 \text{ bits}$$

$$\text{Hidden bytes} = \frac{6,291,456}{8} = 786,432 \text{ bytes}$$

$$\approx 0.75 \text{ MB}$$

**Answer**

Using 1 bit per pixel, approximately 0.75 MB can be hidden.

# Numerical 4: RGB LSB Capacity

An RGB image has $2048 \times 3072$ pixels. If 1 LSB is used in each color channel, calculate the capacity.

$$\text{Pixels} = 2048 \times 3072 = 6,291,456$$

Each pixel has 3 channels: R, G, B.

$$\text{Bits} = 6,291,456 \times 3 = 18,874,368$$

$$\text{Bytes} = \frac{18,874,368}{8} = 2,359,296$$

$$\approx 2.25 \text{ MB}$$

**Answer**

Approximately 2.25 MB can be hidden using 1 LSB per RGB channel.

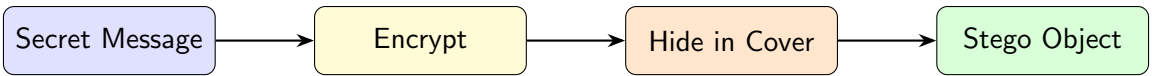
# Advantages of Steganography

- Hides the existence of secret communication.
- Useful when encrypted traffic may attract attention.
- Can be combined with cryptography.
- Works with many types of cover media.
- Useful for watermarking and covert communication.

# Limitations of Steganography

- Often requires large cover files to hide small messages.
- If the hiding method is discovered, the scheme may become useless.
- Compression or editing may destroy hidden information.
- Detection tools may identify abnormal statistical patterns.
- Key distribution may still be required for secure embedding and extraction.

# Cryptography + Steganography




**Best Practice**

Encrypt the message first, then hide the ciphertext inside a cover object.

# Comparison: Cryptography vs Steganography

| **Feature** | **Cryptography** | **Steganography** |
|---|---|---|
| Main goal | Hide meaning | Hide existence |
| Output | Looks scrambled | Looks ordinary |
| Primary risk | Cipher may be broken | Hidden message may be detected |
| Typical method | Encryption algorithm | Embedding method |
| Best use | Confidentiality | Covert communication |

# Class Activity 1

A machine uses 4 rotors. Each rotor has 26 positions.

1. Find the period of the machine.
2. Explain why it is stronger than a single-rotor machine.

**Hint**

$$\text{Period} = 26^r$$

# Class Activity 1: Answer

$$\text{Period} = 26^4$$

$$26^4 = 456,976$$

- A single rotor repeats after 26 letters.
- A 4-rotor machine repeats after 456,976 letters.
- The longer period makes frequency-based cryptanalysis much harder.

# Class Activity 2

A secret message is hidden in the least significant bits of a $1000 \times 1000$ grayscale image.

1. How many bits can be hidden if 1 bit is used per pixel?
2. Convert the capacity into bytes.

# Class Activity 2: Answer

$$\text{Pixels} = 1000 \times 1000 = 1,000,000$$

$$\text{Capacity} = 1,000,000 \text{ bits}$$

$$\text{Bytes} = \frac{1,000,000}{8} = 125,000 \text{ bytes}$$

**Answer**

The image can hide 1,000,000 bits, or 125,000 bytes.

# Quick Quiz

1. What is the purpose of multiple stages of encryption?
2. What is a rotor machine?
3. Why does rotor rotation create polyalphabetic substitution?
4. Calculate the period of a 3-rotor machine.
5. How is steganography different from cryptography?
6. Name two classical steganographic techniques.
7. What is LSB image steganography?
8. Why should encryption and steganography be combined?

## Your Feedback Matters

Please scan the QR code and share your feedback about today's lecture.

- Clarity of concepts
- Quality of examples
- Pace of the lecture
- Suggestions for improvement

**Thank you for your valuable response.**



**Scan to Submit Feedback**

# References


- William Stallings, *Cryptography and Network Security: Principles and Practice*.
- William Stallings and Lawrie Brown, *Computer Security: Principles and Practice*.
- David Kahn, *The Codebreakers*.
- Douglas R. Stinson, *Cryptography: Theory and Practice*.

# Thank You

Questions and Discussion